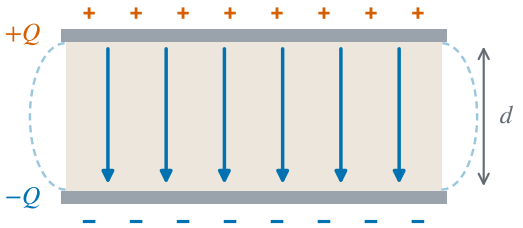
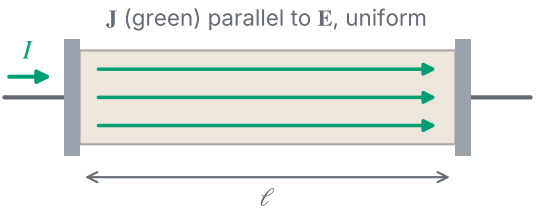
Capacitor: $C = Q/V$



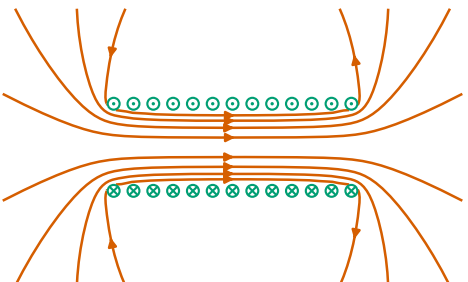
$$Q \to D = \frac{Q}{A_c} \to E = \frac{D}{\varepsilon} \to V = Ed \;\Rightarrow\; C = \frac{\varepsilon A_c}{d}$$

Resistor: $R = V/I$



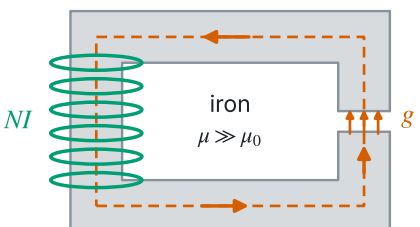
$$I \to J = \frac{I}{A_c} \to E = \frac{J}{\sigma} \to V = E\ell \;\Rightarrow\; R = \frac{\ell}{\sigma A_c}$$

Inductor: $L = \lambda/I$



$$I \to B = \frac{\mu N I}{\ell} \to \Phi = BA_c \to \lambda = N\Phi \;\Rightarrow\; L = \frac{\mu N^2 A_c}{\ell}$$

Magnetic circuit: $\mathcal{R} = \mathcal{F}/\Phi$



$$NI = H_y \ell_y + H_g g, \text{ same } \Phi \;\Rightarrow\; \mathcal{R} = \frac{\ell_y}{\mu A_c} + \frac{g}{\mu_0 A_c}$$